GABRIEL BÉNA

Postdoctoral Researcher @ Institute of Neuroinformatics, UZH / ETH Zurich
PhD @ Imperial College London

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PERSONAL STATEMENT

I am a postdoctoral researcher at the Institute of Neuroinformatics in Zurich, working at the intersection of computational neuroscience, machine learning and neuromorphic engineering. My work asks how complex, adaptive computation can emerge from simple local parts, and how physical constraints, energy, space and device variability, shape the structures that emerge. I have deployed neural networks on real neuromorphic silicon, and I care about the point where theory has to survive contact with hardware.

EXPERIENCE

Postdoctoral Researcher

Institute of Neuroinformatics, UZH / ETH Zurich

May 2026 – present   Zurich, Switzerland

Research in Melika Payvand's group on self-organisation and compositional learning on memristive neuromorphic substrates.

- Local learned rules that grow, repair and continually adapt computation on physical substrates.
- Spatially-embedded networks under energy and locality constraints, and the modularity they produce.

Neuromorphic Researcher & Developer

SpiNNcloud

September 2023 – November 2025   Dresden, Germany

Full-time to October 2024, taken as a one-year interruption of the PhD, then part-time to completion.

- Designed and deployed deep SNNs on neuromorphic chips, using hybrid approaches to leverage the best of conventional and neuromorphic computing.
- First implementation of event-based backpropagation on the SpiNNaker2 platform.
- Active software development and improvement of py-spinnaker2: a high level (**python**) code interfacing with low-level (**C**) applications through middle-layer experiment runner (**C++**) to enable researchers to use Spinnaker2 chips.

PhD Candidate

Imperial College London

January 2021 – March 2026   London, UK

PhD in the Neural Reckoning Group under the supervision of Dan Goodman. Thesis: *Physical constraints and functional demands shape modular neuromorphic intelligence*. Viva voce March 2026.

- Showed that structural modularity alone does not buy functional specialisation: the two align only under the right resource constraints and input statistics.
- Trained continuous Neural Cellular Automata as a universal computational medium.
- Introduced Self-Organising Digital Circuits: local message passing that grows and repairs Boolean logic with no global backpropagation at deployment.

KEY SKILLS

- Proficient in Python, C, Java
- Supervised / Unsupervised Learning, RL
- Expert knowledge of **ANNs** / **SNNs**
- **Pytorch**, **JAX**
- Nice guy

EDUCATION

Operational Research master

Université Paul Sabatier

2019 – 2020   Toulouse, France

Master in Operations Research combining applied Mathematics, Informatics and Engineering aiming to provide strong optimization, modelization and algorithmic competencies.
Ranked 2nd / 29

Master of science

ISAE-Supaero

2016 – 2020   Toulouse, France

Superior Institute of Space and Aeronautics based in Toulouse, ranked 2nd in terms of academic excellence, European leader in aeronautics and space related studies. **Specialization in Data and Decision science** (SDD) and **Machine Learning** applied to Space Systems Operation and Conception.

MP Preparatory classes

Lycée Henri-IV

2014 – 2016   Paris, France

Two-year undergraduate intensive course in mathematics, physics, and computer science preparing a national examination to the top French "Grandes Écoles". **Ranked 301/4800**.

LANGUAGES

French
English
Spanish

INTERESTS / HOBBIES

- **Sports**: Skiing, Surf, Rugby: Regular practice and very good level.
- **Music**: Saxophone, Guitar.
- **Travels**: Morocco (5 years), England (4 years), Japan (6 months), Colombia (6 months).
- **Solarpunk**

ML / Neuromorphic Intern

Prophesee

June 2020 – December 2020 Paris, France

- Trained deep spiking neural networks for event-based cameras, spiking auto-encoders for denoising and compressing tasks
- Worked on end-to-end event-based systems using Intel's LOIHI processor and Prophesee's event-based cameras.

SELECTED PUBLICATIONS

Journal Articles

- **G. Béna** and D. F. M. Goodman, "Dynamics of specialization in neural modules under resource constraints," *Nature Communications*, vol. 16, p. 187, 2025. DOI: 10.1038/s41467-024-55188-9
- M. Ghosh, K. G. Habashy, F. De Santis, **G. Béna**, et al., "Spiking neural network models of interaural time difference extraction via a massively collaborative process," *eNeuro*, vol. 12, no. 7, 2025. DOI: 10.1523/ENEURO.0383-24.2025
- M. Ghosh, **G. Béna**, V. Bormuth, and D. F. M. Goodman, "Nonlinear fusion is optimal for a wide class of multisensory tasks," *PLOS Computational Biology*, vol. 20, no. 7, e1012246, 2024. DOI: 10.1371/journal.pcbi.1012246

Conference Proceedings

- M. Barylli, **G. Béna**, A. Mordvintsev, E. Nisioti, and S. Risi, "Self-organising digital circuits," in *Artificial Life Conference (ALIFE)*, In press; equal contribution, 2026.
- **G. Béna**, M. Faldor, D. F. M. Goodman, and A. Cully, "A path to universal neural cellular automata," in *Genetic and Evolutionary Computation Conference (GECCO Companion)*, 2025. DOI: 10.1145/3712255.3734310
- **G. Béna**, T. Wunderlich, M. Akl, B. Vogginger, C. Mayr, and H. A. Gonzalez, "Event-based backpropagation on the neuromorphic platform SpiNNaker2," in *Neuro-Inspired Computational Elements Conference (NICE)*, 2025. DOI: 10.48550/arXiv.2412.15021